Blinding Lights: Enhancing Object Classification in High-Light Reflection and Overexposed Scenarios

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Abstract

Detecting objects in images with high brightness or reflective surfaces poses significant challenges for computer vision models. In this study, we developed two novel convolutional neural network (CNN) architectures: ResNet+, incorporating channel-wise attention mechanisms, and BrightNet, featuring spatial attention across all residual blocks. These models were evaluated against a baseline ResNet-18 on a subset of the WildCam dataset focusing on overexposed and reflective traffic camera images. While BrightNet demonstrated slight improvements in accuracy for high-brightness images, no significant increase in F1-score was observed across models due to class imbalance and prediction biases. This work contributes novel CNN architectures tailored for challenging lighting conditions and provides insights into the impact of attention mechanisms on object detection performance.

1. Introduction

a. **Background and Motivation**:

Object detection is a cornerstone of computer vision, with applications spanning autonomous vehicles, surveillance systems, and wildlife monitoring. Accurate detection in real-world scenarios is crucial for decision-making and safety. However, bright environments with high reflections and varying contrast (e.g. Figure 1) pose significant challenges to object detection systems. Such conditions often result in feature distortions, reduced model accuracy, and false detections. These inaccuracies could be the difference between life and death, especially in scenarios such as self-driving cars.



Figure 1: Sun glaring on car (Image Source)

Existing methods, including traditional techniques like edge detection and histogram-based approaches, and modern convolutional neural networks (CNNs), struggle to handle overexposed or reflective images effectively. Traditional approaches fail to adapt to dynamic lighting conditions, while CNN-based models often prioritize features poorly, leading to suboptimal performance. These limitations highlight the need for specialized solutions that address the unique challenges of high-brightness and reflective environments, paving the way for this study's focus on developing enhanced CNN architectures.

b. **Project Goals**:

The primary goal of this project is to develop and evaluate convolutional neural network (CNN) architectures specifically tailored to handle challenging bright image conditions, such as those with high reflections and overexposure. To achieve this, the project aims to:

1. Prepare and augment the dataset to filter images in challenging lighting conditions from

the WildCam test set

2. Design and implement enhancements to baseline CNN architectures (ResNet), focusing on extracting meaningful features from bright and reflective images (such as Figure 2)
3. Explore the integration of spatial and channel-wise attention mechanisms to improve feature prioritization and detection accuracy in these conditions.
4. Compare the performance of the proposed architectures against ResNet, providing insights into the effectiveness of attention mechanisms for object detection under extreme lighting challenges.



Figure 2: Sample High Exposure Images taken on WildCam

2. Related Work

Early object detection methods, such as Histogram of Oriented Gradients (HOG) and Scale-Invariant Feature Transform (SIFT), relied on handcrafted features to identify objects in images. While these techniques were effective in controlled lighting and simple backgrounds, they often fail under bright or reflective conditions. Bright environments introduce feature distortions and saturation, making traditional methods inadequate for dynamic and high-contrast scenes.

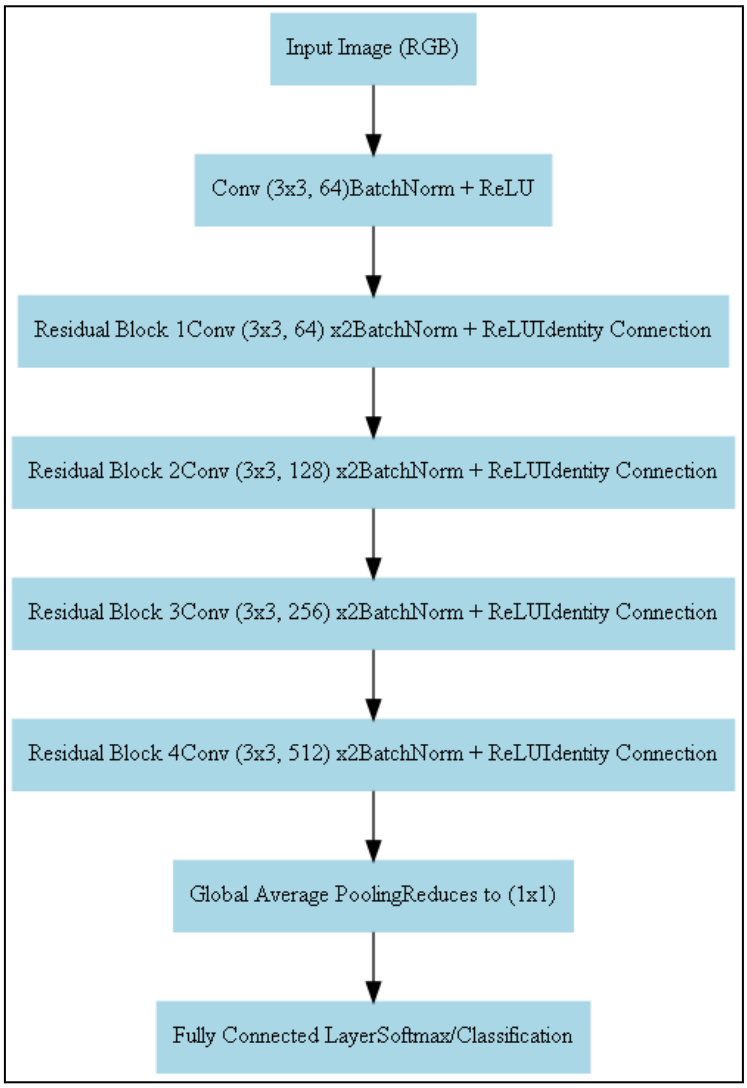
CNNs have revolutionized object detection by automating feature extraction and enabling more robust performance in complex scenarios. Architectures like ResNet, YOLO, and Faster R-CNN have demonstrated significant advancements in detection accuracy and speed. ResNet, in particular, introduced “residual learning frameworks” that alleviate the vanishing gradient problem, allowing for “deeper networks” and improved feature representation (He, 2016). However, these models still struggle in extreme lighting conditions, as standard convolutional layers lack mechanisms to prioritize critical features in reflective or overexposed regions.

Attention mechanisms enhance CNN performance by helping models focus on critical features. Channel-wise attention, like “Squeeze-and-Excitation” (SE) blocks, recalibrates the importance of each feature map channel, emphasizing relevant information while suppressing noise (Hu, 2018). Spatial attention highlights important regions within an image, enabling the model to isolate meaningful patterns even in reflective or overexposed areas. Hybrid approaches combining spatial and channel-wise attention further refine feature prioritization, while multi-scale attention improves detection across varying object sizes. These techniques form the basis for the advanced architectures, ResNet+ and BrightNet, developed in this project.

3. Design and Implementation

a. Model Architectures

ResNet-18: ResNet-18, outlined in He's paper, serves as the baseline model in our experiments. It employs the residual learning framework, which helps mitigate the vanishing gradient problem in deep networks. ResNet-18 consists of a series of basic residual blocks, each with two convolutional layers followed by batch normalization and ReLU activation. These blocks also include shortcut connections that allow gradients to flow more effectively during backpropagation. The simplicity of ResNet-18 makes it a reliable baseline for comparison.



a

Figure 3: ResNet-18 Architecture

ResNet+: ResNet+, the first iteration of this project's novel model, builds upon ResNet-18 by integrating Squeeze-and-Excitation (SE) blocks into its architecture. SE blocks are designed to recalibrate feature channels by emphasizing those deemed most important for the task. This is achieved through global average pooling followed by two fully connected layers that produce channel-wise weights, which are applied to the feature map. While SE blocks add minimal computational overhead, their channel-wise focus provides additional flexibility in feature representation.

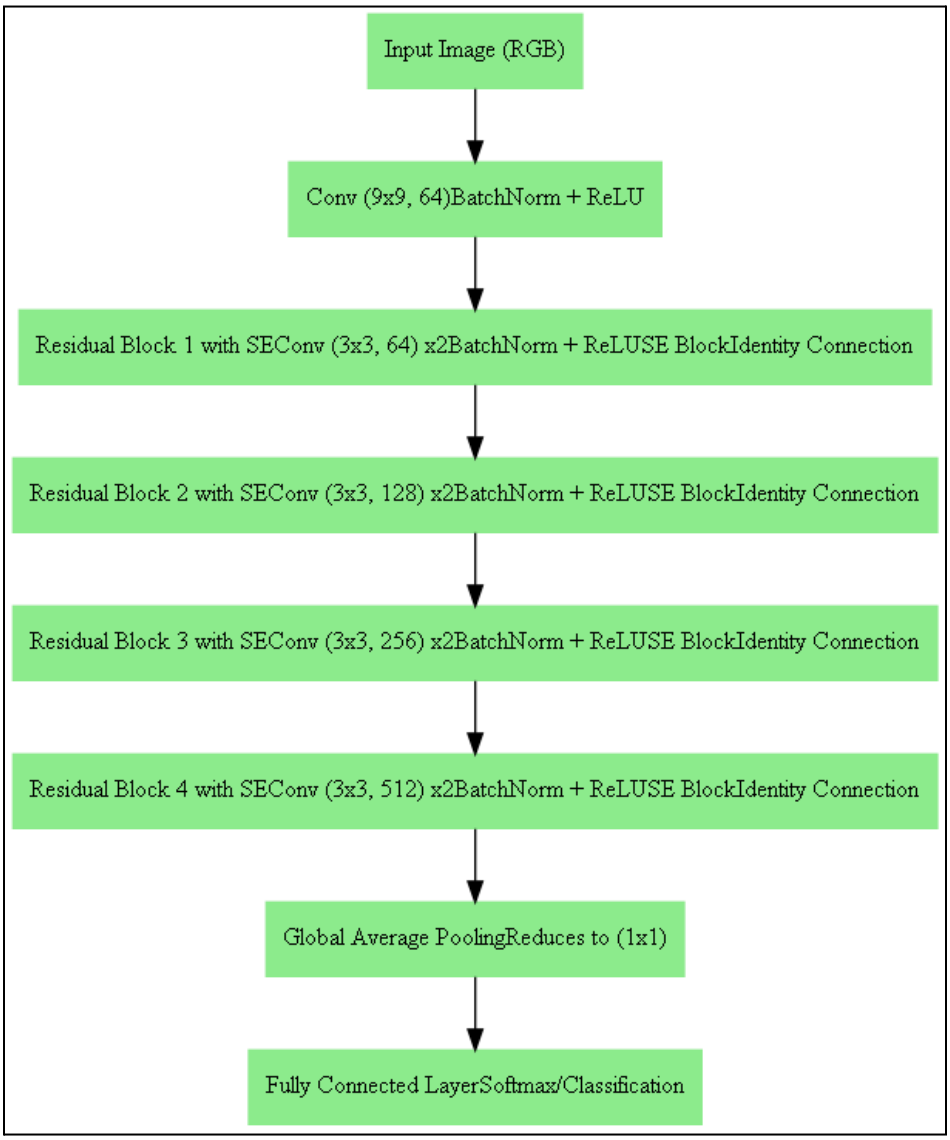


Figure 4: ResNet+ Architecture

BrightNet: BrightNet, the second iteration of this project's model, introduces the BrightFeatureBlock, which is tailored for challenging lighting conditions. Each BrightFeatureBlock consists of two convolutional layers with batch normalization and ReLU activation. It incorporates contrast-focused processing specialized to refine features in bright or high-reflection regions. Additionally, it incorporates a spatial attention mechanism with a 9x9 convolution followed by a sigmoid activation that generates attention maps that highlight the most relevant regions of the image. These maps are applied to the feature maps, allowing BrightNet to focus on informative spatial areas and suppress distractions like glare or reflections.

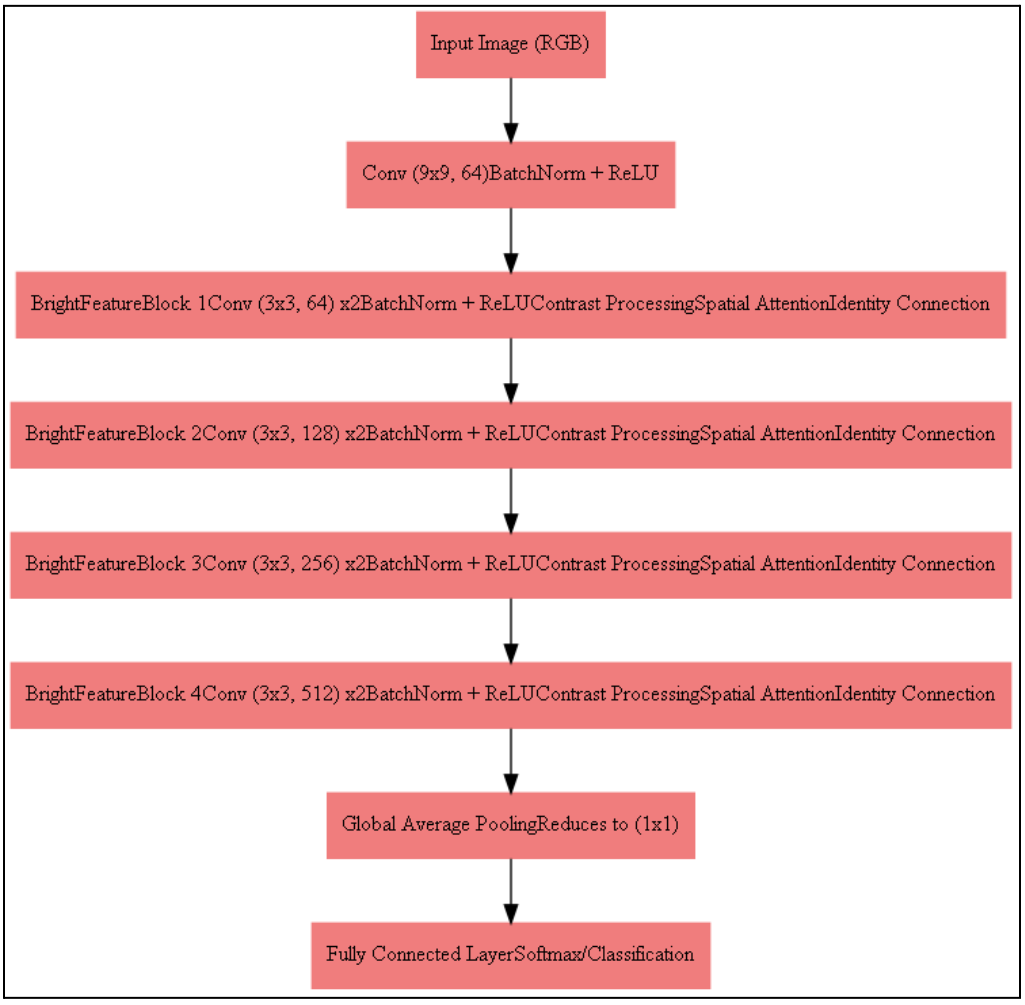


Figure 5: BrightNet Architecture

b. Dataset Preparation

WildCamDataset: This dataset comprises real-world images captured in diverse environments, often featuring challenging lighting conditions. It includes images with reflections, glare, and bright spots, making it ideal for training the models.

BrightCamSubset: While the base WildCamDatset contains bright images, it only accounts for a small proportion of the total dataset. In order to properly evaluate our models on bright images, we grabbed a subset of the Test group in WildCam. We did this by filtering out images below a certain brightness threshold. At first, a large portion of the filtered images belonged to the Rabbit class, so in order to have an equal class spread for testing, we limited the number of images to around 500 for each class. This gave us a great testing set for the models.



Figure 6: Sample Images in BrightCamSubset

	Rabbit	Bobcat	Cat
Test Set	2755	2209	1737
Bright Set	551	442	348
Percentage of Data	41%	33%	26%

Table 1: Dataset Composition

One issue, highlighted by Table 1, is the class composition of the datasets. This has possible implications that may be affecting the results of this study. The over-representation of rabbits compared to bobcats and cats could lead to biased models that perform better on the more represented class (rabbits). Underrepresentation of bobcats and cats in the dataset can result in poor model performance for these classes. We will see later in our confusion matrix analysis that this over and under representation of different classes shines through during model performance.

4. Results

In this section, we provide a comprehensive evaluation of the three models—ResNet-18, ResNet+, and BrightNet—on the WildCam dataset. The dataset includes the Test Set and Bright Set designed to test the models under different lighting conditions, from normal images to those with high light reflection. The evaluation metrics include Precision, Recall, F1-Score, and Accuracy, offering a well-rounded view of model performance in object detection tasks. Additionally, we will conduct visual comparisons of the models' predictions on test images, analyzing feature maps and object detection results.

a. Quantitative Results

i. Training Loss Across Iterations

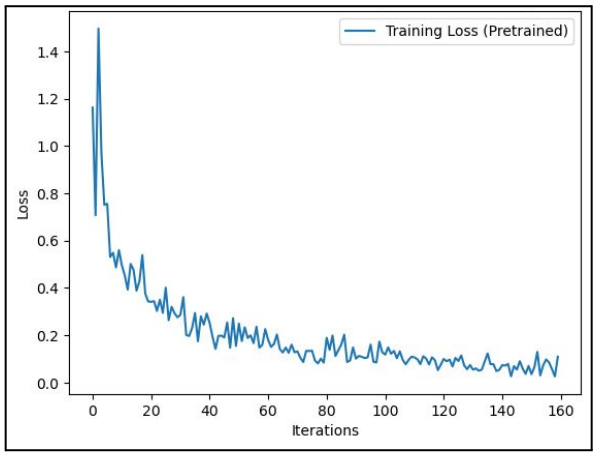


Figure 7: ResNet-18 Training Loss

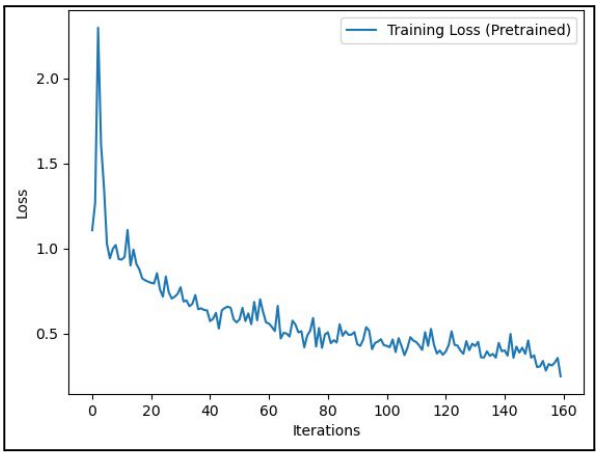


Figure 8: ResNet+ Training Loss

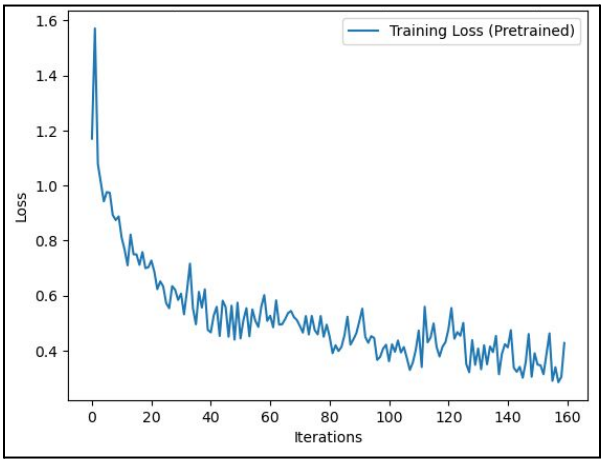


Figure 9: BrightNet Training Loss

The ResNet-18 model starts with a high loss around 1.2, which is expected for the initial phase of training where the model's weights are randomly initialized. The loss continues to decrease gradually after the initial sharp drop and eventually stabilizes around a very low value close to 0.1 by iteration 160, suggesting that the model has converged and is no longer making significant adjustments.

ResNet+ starts with a high loss around 2.3, potentially due to the additional complexity introduced by the SE blocks. The loss begins to decrease but with more fluctuations compared to ResNet-18. It stabilizes around 0.5, higher than ResNet-18's final loss. Near the end it starts to drop slightly more, suggesting that with more training epochs a lower final loss could be achieved.

BrightNet starts with a medium loss around 1.5. The loss continues to decrease gradually, with fewer fluctuations compared to ResNet+. It stabilizes around 0.4, slightly higher than ResNet-18 but lower than ResNet+. Similarly to ResNet+, it appears as though a lower training loss may be achievable given more training epochs.

All three models show a rapid initial decrease in training loss, indicating effective early learning. ResNet-18 and BrightNet start with slightly lower initial losses compared to ResNet+. ResNet-18 achieves the lowest final loss, followed by BrightNet, with ResNet+ having the highest. This order reflects the different complexities and specializations of the models:

- **ResNet-18:** Simplest architecture, leading to the most stable and lowest final loss.
- **ResNet+:** Additional SE blocks introduce complexity, leading to higher final loss.
- **BrightNet:** Uses special blocks for bright conditions, balancing complexity and stability.

ResNet-18's low final loss suggests it may overfit the training data more compared to the other models, particularly in scenarios where simplicity leads to memorization.

ResNet+'s higher and more fluctuating loss indicates its architecture's added complexity, which may offer better generalization but at the cost of higher training instability.

BrightNet's steadier decrease and moderate final loss reflect its balanced approach, showing robustness in learning and adapting to high brightness conditions without overfitting excessively.

ii. Model Performance

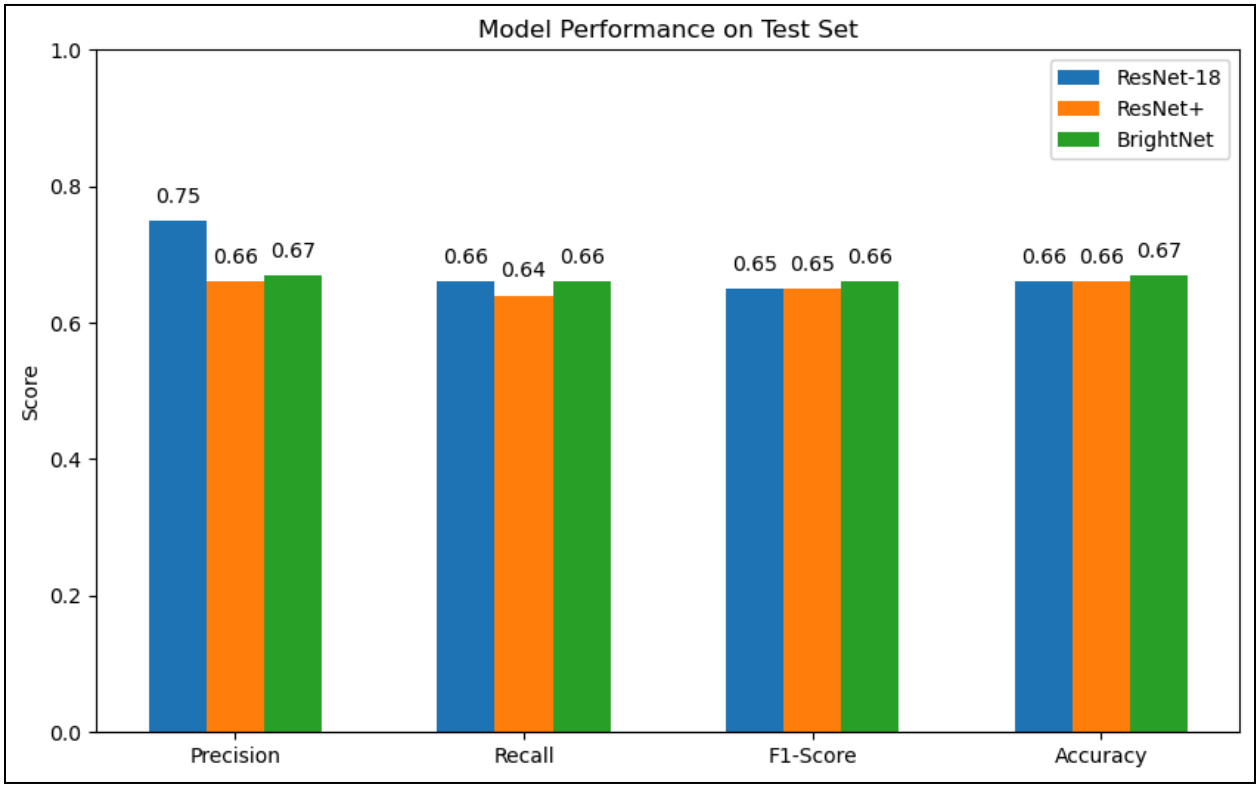


Chart 1: Model Performance on Test Set

ResNet-18 Test Set:

- **Precision (0.75)**: indicates that ResNet-18 is very accurate in identifying true positives, making few false positive errors.
- **Recall** (**0.66**): indicates that ResNet-18 is relatively good at identifying most true positives.
- **F1-Score** (**0.65)**: indicates that ResNet-18 has a fair balance between avoiding false positives and capturing true positives.
- **Accuracy (0.66)**: indicated ResNet-18 a reliable baseline with consistent performance.

ResNet+ Test Set:

- **Precision (0.66)**: lower precision compared to ResNet-18 suggests that the additional complexity of SE blocks might introduce some noise, leading to a higher rate of false positives.
- **Recall (0.64)**: slightly lower recall than ResNet-18, suggesting that while ResNet+ tries to balance precision and recall, it may miss some true positives due to its complexity.
- **F1-Score (0.65)**: matching ResNet-18, the score suggests that despite its lower precision and recall, ResNet+ achieves a balanced performance.

- **Accuracy (0.66)**: matching ResNet-18, suggesting that despite individual metric variations, ResNet+ achieves an overall similar performance on the test set.

BrightNet Test Set:

- **Precision (0.67)**: moderately lower than ResNet-18, however, BrightNet's precision still suggests its design maintains the ability to mitigate false positives.
- **Recall (0.66)**: matching ResNet-18, BrightNet's recall indicates it is moderately effective at detecting true positives.
- **F1-Score (0.66)**: slightly higher than ResNet-18, indicating that BrightNet balances precision and recall effectively, leveraging its bright condition enhancements to maintain strong performance overall.
- **Accuracy (0.67)**: a slight edge in accuracy over ResNet-18 indicates that BrightNet's specialized features do not hinder overall performance.

ResNet-18 performs robustly with high precision and balanced recall, making it an effective baseline model. ResNet+, despite its intended architectural improvements, shows a trade-off in precision and recall but achieves balanced performance with a similar accuracy to ResNet-18. ResNet+ introduces additional complexity to the ResNet-18 architecture through its SE blocks. While these enhancements aim to improve feature extraction and model sensitivity, they can sometimes overemphasize noise or amplify irrelevant features, leading to an increased rate of false positives and, consequently, lower precision. Moreover, the added architectural complexity can make optimization more challenging, particularly in scenarios with noisy data, where the model might overfit to dominant patterns at the expense of missing subtler true positives, resulting in slightly lower recall. This trade-off highlights the balance between generalization and over-specialization, as ResNet+ may struggle to equally prioritize all relevant features in the dataset. Despite these challenges, ResNet+ achieves an F1-score and accuracy comparable to ResNet-18, suggesting that while its enhancements introduce some trade-offs, they do not significantly compromise overall performance and may provide benefits in specific scenarios. BrightNet, with its targeted improvements, demonstrates slightly better performance across some metrics, indicating a possible improvement in the model, specifically on brighter images within the Test Set.

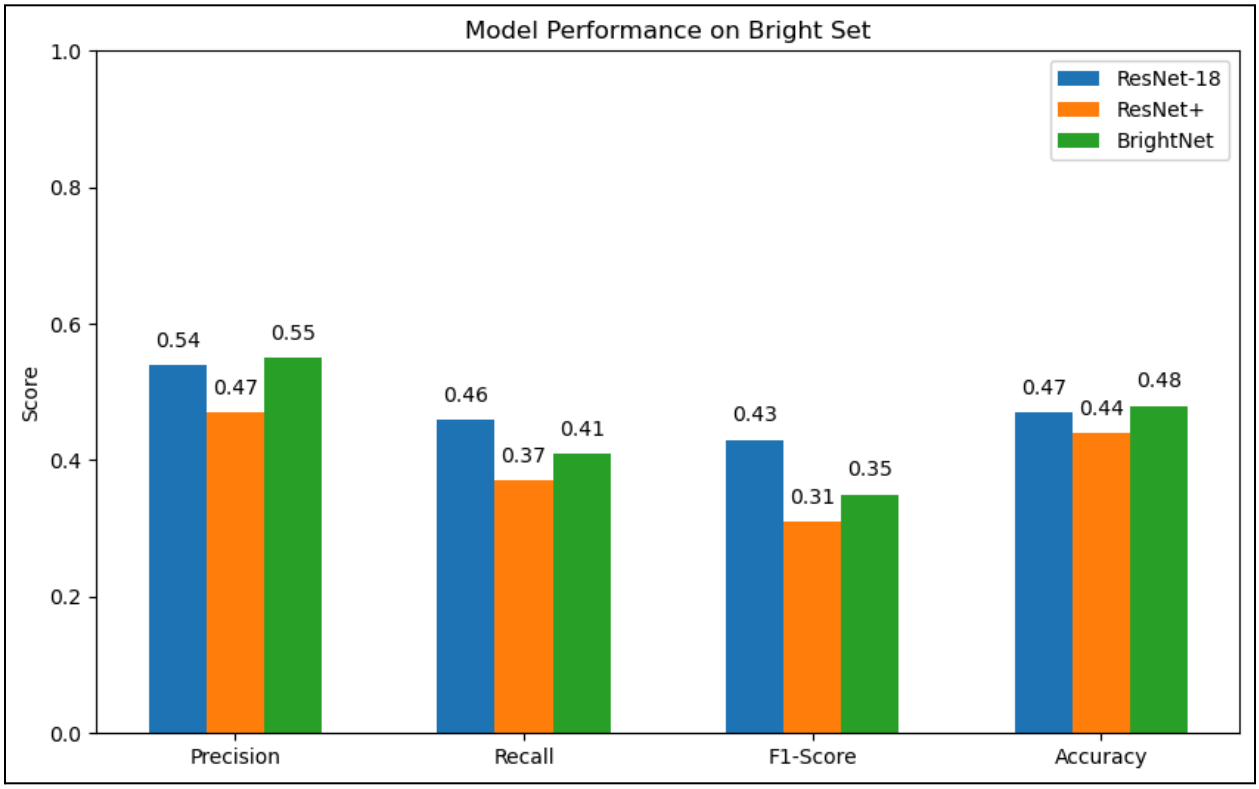


Chart 2: Model Performance on Bright Set

ResNet-18 Bright Set:

- **Precision (0.54)**: indicates ResNet-18 has a moderate ability to correctly identify true positives but also has a significant number of false positives in bright conditions.
- **Recall (0.46)**: indicates ResNet-18 misses over half of the true positives in bright conditions.
- **F1-Score (0.43)**: indicates that while ResNet-18 performs somewhat well in bright conditions, there's a noticeable drop in overall performance compared to standard conditions.
- **Accuracy (0.47)**: indicates that ResNet-18 can correctly predict a little less than half of the cases in bright conditions.

ResNet+ Bright Set:

- **Precision (0.47)**: lowest precision indicates that ResNet+ struggles more with false positives under bright conditions compared to ResNet-18.
- **Recall (0.37)**: lowest recall among the models, indicating that ResNet+ significantly struggles to capture true positives in bright conditions.
- **F1-Score (0.31)**: lowest F1-Score, indicating that ResNet+ finds it hard to balance precision and recall in bright conditions.

- **Accuracy (0.44)**: indicates lower overall prediction accuracy compared to ResNet-18. This aligns with the increased false positives and missed detections.

BrightNet Bright Set:

- **Precision (0.55)**: shows the highest precision, showing its improved ability to correctly identify true positives and manage false positives more effectively in bright conditions.
- **Recall (0.41)**: although not significantly lower, BrightNet's recall is worse than ResNet-18, indicating a lessened ability to detect true positives.
- **F1-Score (0.35)**: slightly better than ResNet+, reflecting a marginally better balance between precision and recall.
- **Accuracy (0.48)**: the slight edge in accuracy indicates BrightNet's improved robustness in bright conditions. Although the improvement is modest, it reflects the benefits of BrightNet's specialized features.

ResNet-18 demonstrates moderate performance in bright conditions but shows a noticeable drop in all metrics compared to standard conditions, indicating the model's limitations in handling high brightness. ResNet+ struggles the most among the models, highlighting that the SE blocks do not provide an advantage in bright conditions. The significant drop in all scores underscores the model's difficulty in accurately detecting objects in bright environments, especially when compared to ResNet-18 and BrightNet. BrightNet shows the best performance across many metrics in bright conditions, though with room for improvement, suggesting its specialized design for bright environments helps reduce the impact of high brightness on performance, making it more robust compared to ResNet-18 and ResNet+. Interestingly, in these bright conditions, precision and F1-score drop while accuracy increases, which can be explained by the interplay of class imbalances and metric calculations. Accuracy can improve if the model correctly classifies the majority class (true negatives), even as precision and F1-score decline due to errors in the minority positive class. High brightness may lead the model to become over-cautious, reducing the number of positive predictions to avoid false positives, which impacts precision and F1-score but allows accuracy to benefit from improved true negative classifications. This highlights how the metrics diverge, emphasizing BrightNet's relative robustness in such challenging conditions.

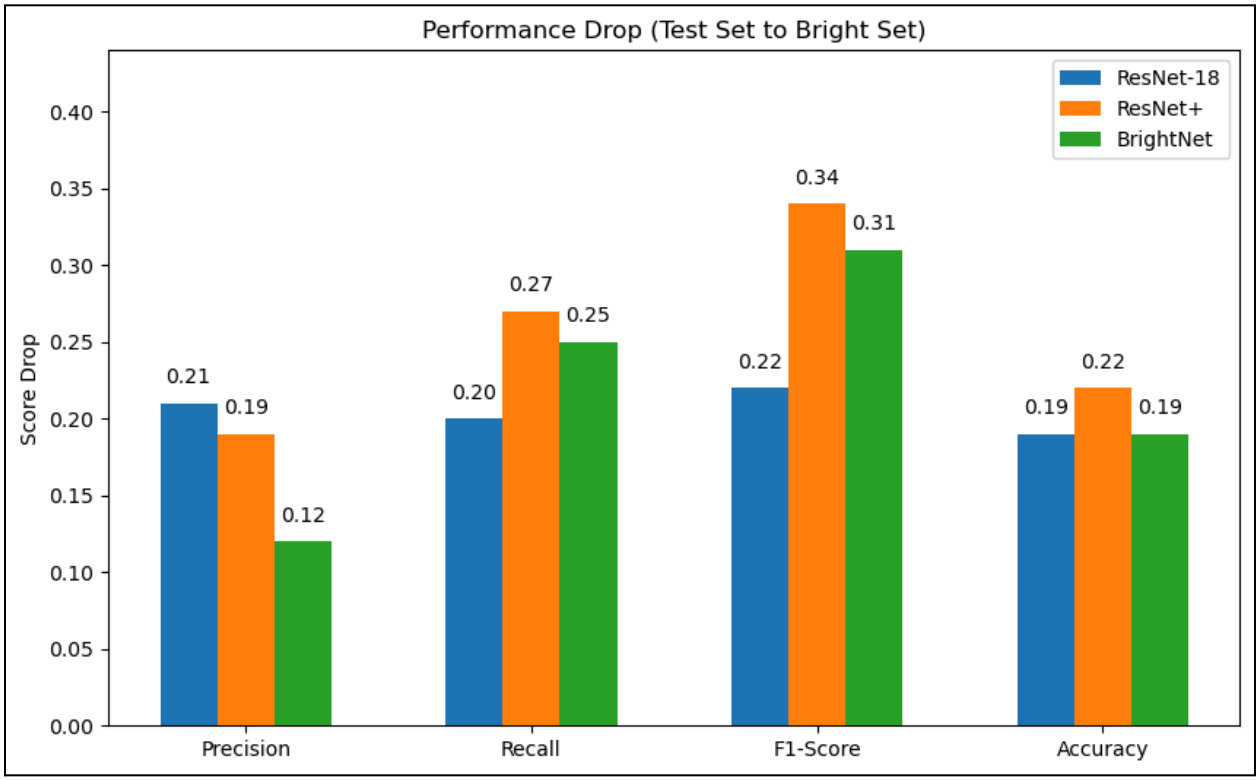


Chart 3: Performance Drop (Test Set to Bright Set)

ResNet-18 Performance Drop:

- **Precision (0.21):** a large drop in precision suggests that ResNet-18 struggles with false positives more in bright conditions
- **Recall (0.20):** a large recall drop indicates that ResNet-18 also misses more true positives under bright conditions.
- **F1-Score (0.22):** a decrease in F1-Score reflects the overall performance degradation.
- **Accuracy (0.19):** the drop in accuracy confirms the overall decline in performance, indicating that ResNet-18's general capability to make correct predictions is worsened in high brightness.

ResNet+ Performance Drop:

- **Precision (0.19):** the precision drop indicates that ResNet+ also struggles with false positives in bright conditions, although slightly less so than ResNet-18.
- **Recall (0.27):** the largest recall drop among the models highlights ResNet+'s severe difficulty in capturing true positives under bright conditions.
- **F1-Score (0.34):** the substantial decrease in F1-Score reflects the combined challenges in precision and recall, highlighting ResNet+'s overall performance degradation.
- **Accuracy (0.22):** a notable drop in accuracy shows that ResNet+ faces significant challenges in maintaining correct predictions under bright conditions.

BrightNet Performance Drop:

- **Precision (0.12):** the smallest drop in precision among the models indicates that BrightNet is better at maintaining accuracy in positive predictions under bright conditions.
- **Recall (0.25):** while BrightNet shows an improvement, the recall drop still indicates a struggle in detecting true positives under bright conditions.
- **F1-Score (0.31):** the F1-Score drop, though significant, is less severe compared to ResNet+, reflecting a better balance between precision and recall under bright conditions.
- **Accuracy (0.19):** the equal drop in accuracy to ResNet-18 indicates that BrightNet's overall capability to make correct predictions is equal to ResNet-18.

ResNet-18 suffers significant drops across all metrics, particularly in precision and recall, indicating it struggles with both false positives and missed detections in bright conditions. This suggests the need for architectural enhancements to better handle such scenarios. ResNet+ shows the largest drop in recall and F1-Score, highlighting its difficulty in maintaining performance under bright conditions. The SE blocks, while beneficial in standard scenarios, do not translate well to high brightness, leading to significant performance degradation. BrightNet exhibits the smallest drops across some metrics, especially in precision and notably equivalent to ResNet-18 in accuracy, indicating its improved ability to handle high brightness.

For per class data on ResNet-18, reference Appendix A: Table 2

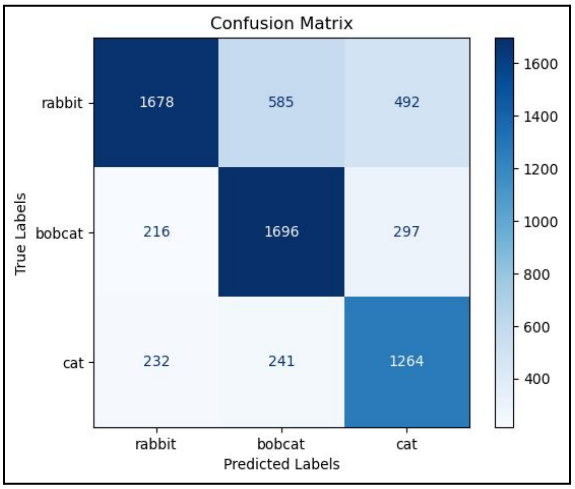
For per class data on ResNet+t, reference Appendix A: Table 3

For per class data on BrightNet, reference Appendix A: Table 4

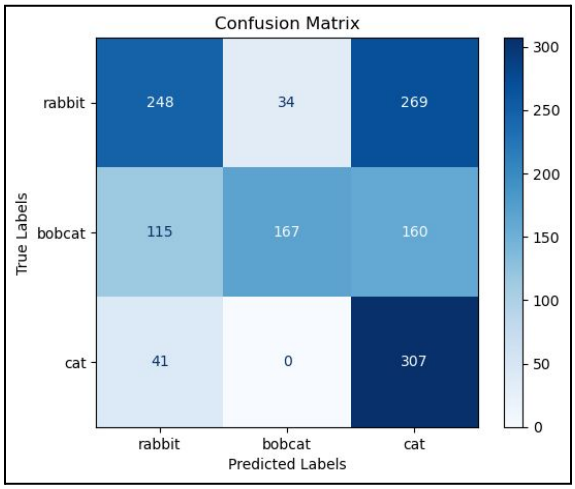
For complete average data, reference Appendix A: Table 5

For model performance on Validation set, reference Appendix A: Chart 4

iii. Confusion Matrices



CM 1: ResNet-18 on Test Set



CM 2: ResNet-18 on Bright Set

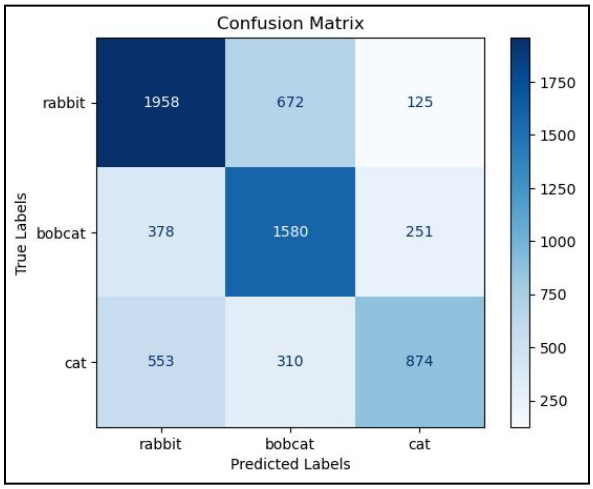
Test Set Performance:

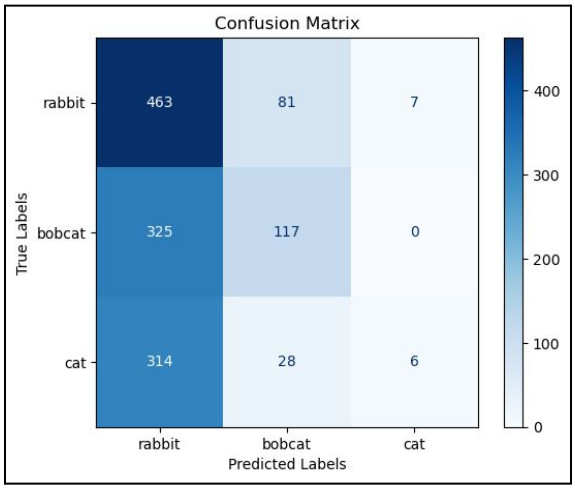
- **Strong Points:** Performs best at identifying bobcats.
- **Weak Points:** Slightly struggles to identify rabbits.

Bright Set Performance:

- **Strong Points:** Performs relatively better with cats under bright conditions.
- **Weak Points:** Struggles with identifying all classes as a cat.

In bright conditions, the model struggles significantly to correctly classify rabbits and bobcats, leading to increased misclassifications primarily as cats. This suggests that high brightness severely impacts the model's ability to distinguish rabbit and bobcat features from cat features. While the model performs relatively better with cats under bright conditions, it still has high false positive rates among the other classes, leading to high rates of misclassification. This suggests that the model is likely not picking up on distinguishable features from rabbits and bobcats and instead hyper focuses on those that would cause it to mistake everything for a cat.





CM 3: ResNet+ on Test Set

CM 4: ResNet+ on Bright Set

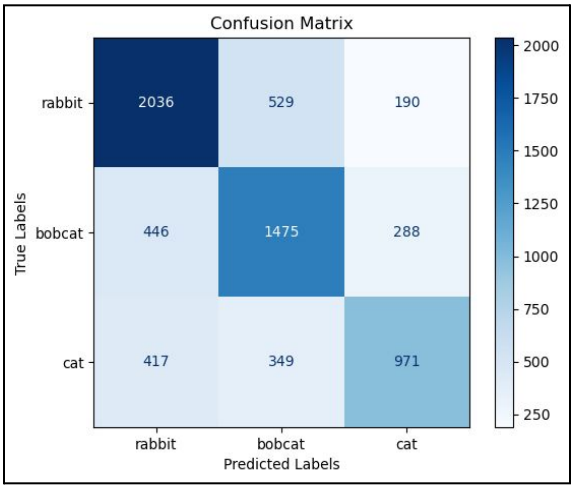
Test Set Performance:

- **Strong Points:** Performs best at identifying rabbits.
- **Weak Points:** Struggles to distinguish between rabbits and cats.

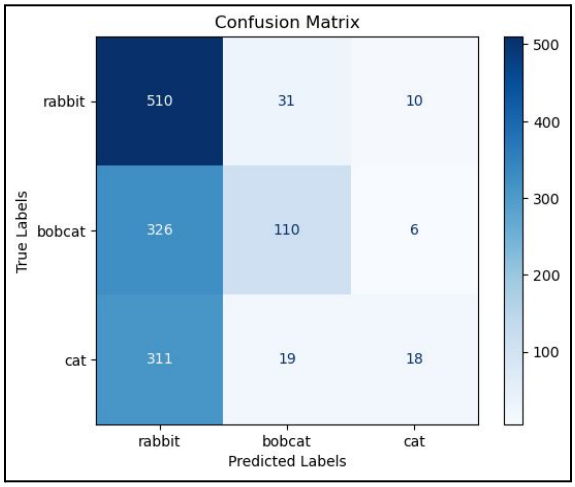
Bright Set Performance:

- **Strong Points:** Still performs best with rabbits.
- **Weak Points:** Struggles with identifying all classes as a rabbit or bobcat.

ResNet+ shows a severe performance degradation at identifying bobcats and cats when moving from the test set to the bright set. The significant drop in true positives and increased misclassifications indicate that the SE blocks must be emphasizing characteristics that would lead to misidentification of these classes as rabbits, as ResNet-18 had a different issue of identifying everything as cats when switching to the bright set.



CM 5: BrightNet on Test Set



CM 6: BrightNet on Bright Set

Test Set Performance:

- **Strong Points:** Performs best at identifying rabbits.
- **Weak Points:** Slightly struggles to distinguish between rabbits and bobcats.

Bright Set Performance:

- **Strong Points:** Still performs best with rabbits.
- **Weak Points:** Struggles with identifying all classes as a rabbit.

BrightNet also experiences degradation when moving from the test set to the bright set. It appears to be having the same issue as ResNet+ when moving from normal to bright conditions, where it begins misclassifying all classes as a rabbit. However, in normal conditions, we see that BrightNet struggles to distinguish between bobcats and rabbits

b. Qualitative Results

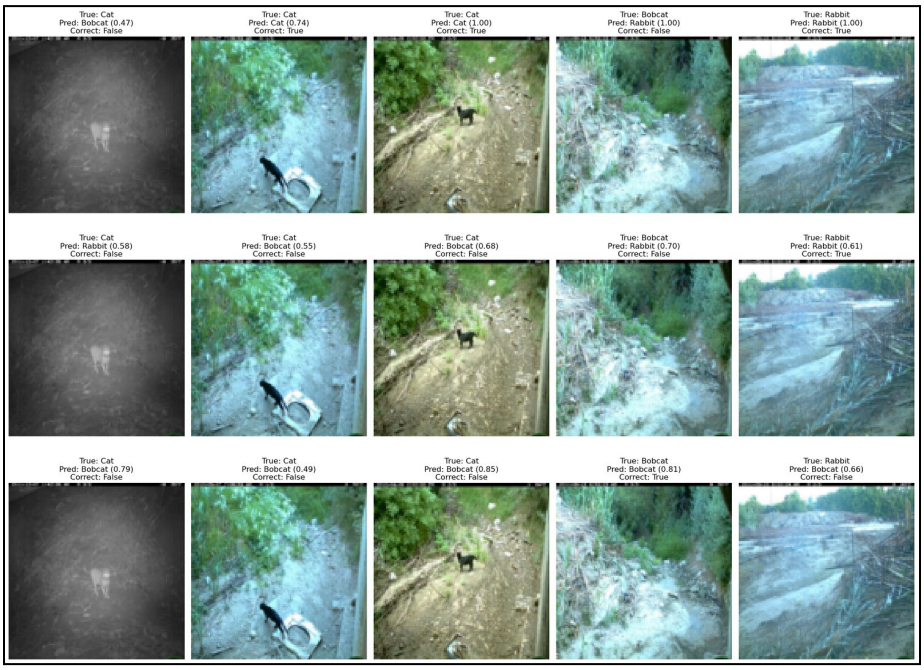


Figure 10: Predictions on Test Set (Top to Bottom ResNet-18, ResNet+, BrightNet)

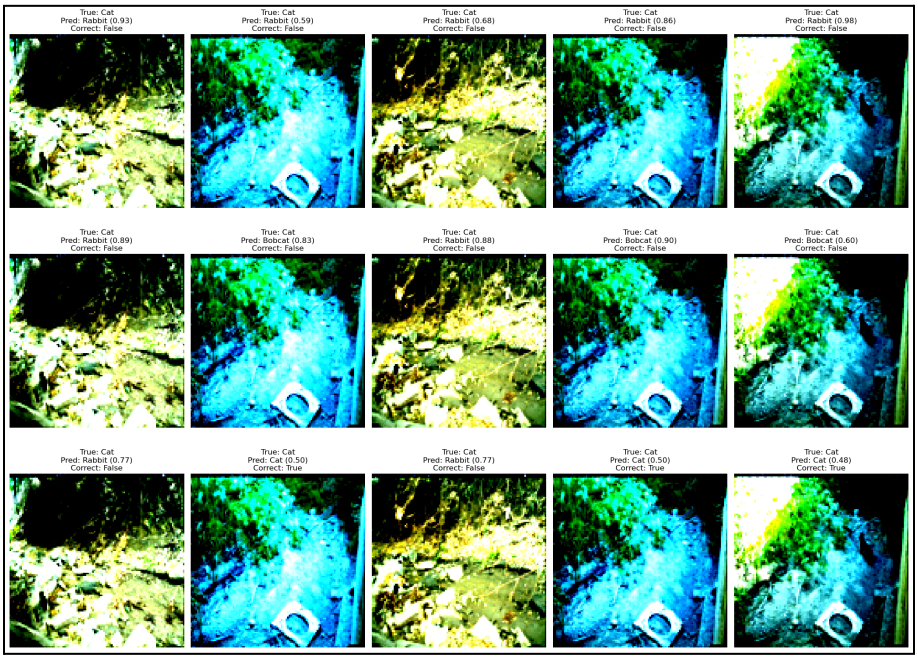


Figure 11: Predictions on Bright Set (Top to Bottom ResNet-18, ResNet+, BrightNet))

For Larger versions of Figures 10, 11 reference Appendix B Figure 10 and Figure 11
For additional predictions, reference Appendix B Figure 14 and Figure 15

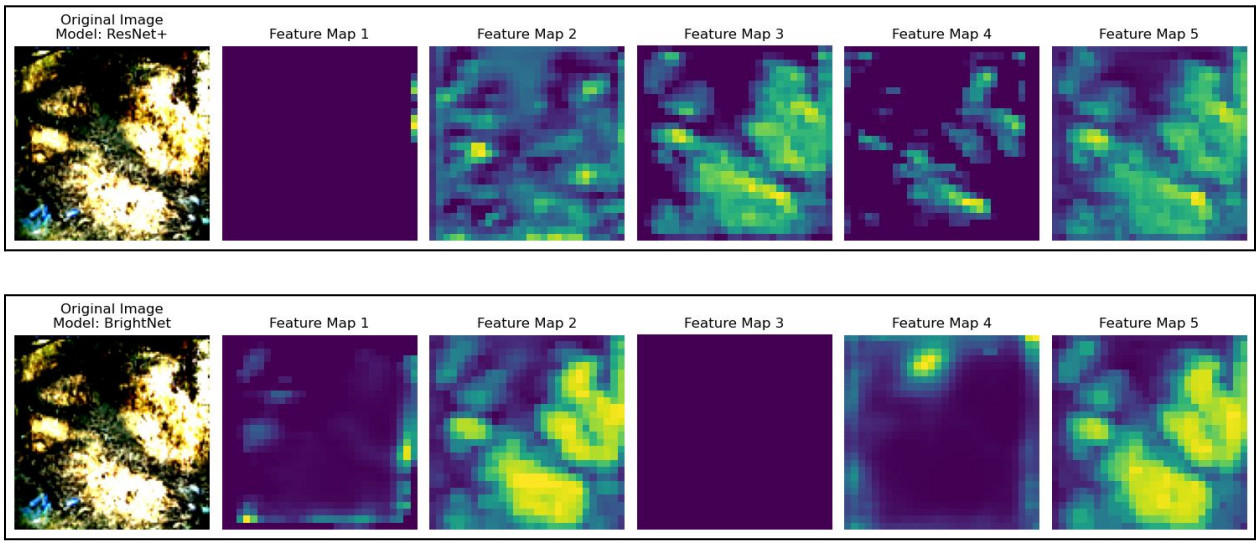


Figure 12: Feature Maps for ResNet+ and BrightNet

The feature maps from ResNet+ reveal significant challenges in handling high-brightness images. The first map shows almost no activation, appearing completely dark, which suggests the model struggles to identify meaningful features in the initial layers. This lack of early activation may indicate that the SE blocks, although designed to recalibrate channel importance, fail to effectively address the extreme brightness present in the image. As the feature maps progress (Maps 2–5), activations appear sporadic and poorly structured, with minimal focus on specific regions of interest. This scattered activation pattern highlights a lack of robustness in feature extraction, where the model seems overwhelmed by the overexposed regions and noise. Consequently, ResNet+ struggles to discern meaningful patterns, reducing its ability to perform effectively under challenging lighting conditions.

BrightNet's first map exhibits visible and meaningful activations, indicating that the model begins extracting relevant features early in the network. Subsequent maps (Maps 2, 4, 5) show progressively more focused and refined activations, with distinct regions of interest being highlighted. This structured response suggests that BrightNet's architectural design effectively mitigates the effects of high brightness, suppressing irrelevant or noisy features while emphasizing key patterns in the image. An interesting observation in BrightNet's feature maps is the apparent lack of activation in Layer 3. BrightNet's architecture is optimized to suppress features influenced by high-brightness noise. The absence of activation in Layer 3 may reflect the model's deliberate effort to reduce the influence of overexposed regions and prioritize features with higher relevance to the detection task.

For additional feature map examples, reference Appendix B: Figure 16 and Figure 17

5. Discussion

The results of this study highlight the varied strengths and weaknesses of ResNet-18, ResNet+, and BrightNet across standard and bright conditions. While ResNet-18 serves as a reliable baseline, ResNet+ and BrightNet introduce architectural additions that aim to improve performance, particularly in challenging bright environments. However, these improvements do not always work, and also come with trade-offs, particularly in terms of complexity.

ResNet-18:

- **Strengths:** ResNet-18 exhibits strong overall performance on the test set, achieving the lowest training loss and demonstrating stable and consistent performance metrics such as Precision (0.75), Recall (0.66), and F1-Score (0.65). The simplicity of its architecture allows it to converge quickly and effectively without significant overfitting.
- **Weaknesses:** Despite its baseline robustness, ResNet-18 struggles significantly under bright conditions. With a substantial drop in Precision (0.21) and Recall (0.20) when transitioning to the Bright Set, the model's ability to handle high brightness is clearly limited. Misclassifications increase, with the model often predicting bright-region objects as cats due to a lack of spatial focus.

ResNet+:

- **Strengths:** The introduction of SE blocks in ResNet+ aims to recalibrate feature channels by weighting their importance dynamically. This architectural enhancement offers modest benefits in balancing Precision and Recall under standard conditions. However, the SE blocks introduce complexity that slightly destabilizes the model during training, leading to fluctuations in loss curves and slower convergence.
- **Weaknesses:** Under bright conditions, ResNet+ suffers the most significant degradation among the three models, with Recall dropping by 0.27 and F1-Score by 0.34. The SE blocks emphasize channel-wise relationships but fail to incorporate spatial awareness, causing the model to misclassify objects, especially under high brightness. This phenomenon is particularly evident in the confusion matrix, where misclassifications as rabbits dominate.

BrightNet:

- **Strengths:** BrightNet's architecture, designed specifically for bright conditions, integrates BrightBlocks with a spatial attention mechanism. This allows the model to focus on relevant regions in bright images, mitigating the impact of reflections and glare. BrightNet demonstrates the most consistent performance in bright conditions, with a small drop in Precision (0.12) compared to ResNet-18 and ResNet+.

- **Weaknesses:** Despite its advantages, BrightNet still struggles with Recall (0.41) under bright conditions, indicating that while it mitigates false positives effectively, it may miss some true positives. Additionally, its spatial attention mechanism, while beneficial, does not completely overcome the challenges posed by bright environments, as seen in its tendency to misclassify objects as rabbits under these conditions.

The Impact of SE Blocks on ResNet+

SE blocks in ResNet+ recalibrate feature channels using a squeeze-and-excitation mechanism. While this approach improves channel-level representation, it lacks the spatial awareness needed to address brightness-specific challenges. In bright environments, SE blocks may overemphasize irrelevant features, leading to higher false positive rates and poor recall. This limitation is evident in ResNet+'s performance on the Bright Set, where it exhibits the highest degradation among all models.

The Role of BrightBlocks and Spatial Awareness in BrightNet

BrightNet addresses the spatial limitations of SE blocks by introducing BrightBlocks, which combine contrast-focused processing with spatial attention. The spatial attention mechanism enables BrightNet to highlight relevant regions in an image, effectively suppressing distractions caused by glare and reflections. This design choice, inspired by Hu's 2018 paper on attention, is reflected in BrightNet's higher precision and steadier performance across metrics in bright conditions. However, the model's Recall drop indicates that its spatial focus might still miss certain true positives, suggesting room for improvement in balancing precision and recall. Additionally, BrightNet's misclassification patterns, particularly its tendency to predict objects as rabbits, highlight the need for further refinement of its spatial attention mechanisms.

In summary, **BrightNet marginally outperforms ResNet-18**, particularly on the Bright Set, showcasing its advantage in handling bright image scenarios. In contrast, ResNet-18 and ResNet+ perform similarly across the Test Set, with ResNet-18 achieving slightly higher accuracy. ResNet+, while capable on balanced datasets, exhibits slightly lower recall and precision overall. However, ResNet+ performs relatively poorly on the Bright Set, displaying much lower accuracy compared to both ResNet-18 and BrightNet. These findings highlight BrightNet's potential while underscoring the need for further refinement for high-brightness conditions.

6. Future Work

Future iterations of this project should aim to enhance model performance by integrating multi-scale attention mechanisms, which can capture features at varying resolutions for improved detection. Combining SE blocks with spatial attention offers a promising hybrid approach to focus on both critical regions and channels, addressing the challenges posed by reflective and overexposed conditions. Additionally, further hyperparameter tuning, such as adjusting learning rates and batch sizes, could unlock additional performance gains.

Expanding the dataset is another crucial step for improving robustness. Collecting more real-world images with diverse lighting conditions and introducing augmented data, such as simulated glare or bright objects, would help models generalize better. Leveraging larger datasets like ImageNet, YOLO, or COCO for fine-tuning could further enhance detection capabilities and provide a solid foundation for tackling extreme brightness scenarios. This would also allow for the project to expand to images of cars, cycles, pedestrians, etc., so that the findings can be applied in the real-world, such as for self-driving detection algorithms.

The models also need to be tested on entirely new datasets, such as those featuring low-light or extreme brightness conditions, to evaluate generalization. Deploying the models in real-world applications like wildlife detection will help assess their robustness in practical scenarios. Optimizing BrightNet for edge devices through pruning and quantization techniques will ensure computational efficiency, enabling reliable performance in resource-constrained environments.



Figure 13: ImageNet (Image Source)

7. Conclusion

This project tackled the challenge of object detection in bright and reflective conditions by developing and comparing novel CNN architectures, BrightNet and ResNet+, with the baseline ResNet-18. While BrightNet showed marginal improvements in accuracy for both overall and brighter images, ResNet+ underperformed in comparison. However, the reliability of BrightNet's results was limited due to its tendency to overpredict the dominant class, 'rabbit,' in the WildCam dataset.

Despite these limitations, the project provided valuable insights into the role of spatial and channel-wise attention mechanisms for handling challenging lighting conditions. The findings highlight the need for better class balancing and further refinement of attention mechanisms to achieve robust and reliable performance.

Future iterations of this work could explore enhanced dataset diversity, refined attention strategies, and improved model evaluation techniques to ensure better generalization and applicability in real-world scenarios. As discussed, developing models that excel on overexposed images is crucial and has the potential to fundamentally transform the implementation of algorithms in real-world applications, ensuring greater reliability and effectiveness in diverse and challenging environments.

References

He, Kaiming, et al. "Deep residual learning for image recognition." *Proceedings of the IEEE conference on computer vision and pattern recognition*. 2016.

Hu, Jie, Li Shen, and Gang Sun. "Squeeze-and-excitation networks." *Proceedings of the IEEE conference on computer vision and pattern recognition*. 2018.

Appendix A

ResNet-18 Performance				
	Precision	**Recall**	**F1-Score**	**Support**
Test Set				
Class 0 (Rabbit)	0.90	0.52	0.66	2755
Class 1 (Bobcat)	0.51	0.93	0.66	2209
Class 2 (Cat)	0.83	0.52	0.64	1737
Validation Set				
Class 0 (Rabbit)	0.94	0.95	0.94	1003
Class 1 (Bobcat)	0.95	0.93	0.94	900
Class 2 (Cat)	0.94	0.95	0.94	791
Bright Set				
Class 0 (Rabbit)	0.63	0.32	0.42	551
Class 1 (Bobcat)	0.40	0.85	0.55	442
Class 2 (Cat)	0.60	0.23	0.33	348

Table 2: ResNet-18 Per Class Performance

ResNet+ Performance				
	Precision	**Recall**	**F1-Score**	**Support**
Test Set				
Class 0 (Rabbit)	0.68	0.71	0.69	2755
Class 1 (Bobcat)	0.62	0.72	0.66	2209
Class 2 (Cat)	0.70	0.50	0.59	1737
Validation Set				
Class 0 (Rabbit)	0.79	0.93	0.85	1003
Class 1 (Bobcat)	0.92	0.89	0.90	900
Class 2 (Cat)	0.88	0.72	0.80	791
Bright Set				
Class 0 (Rabbit)	0.42	0.84	0.56	551
Class 1 (Bobcat)	0.52	0.26	0.35	442
Class 2 (Cat)	0.46	0.02	0.03	348

Table 3: ResNet+ Per Class Performance

BrightNet Performance				
	Precision	**Recall**	**F1-Score**	**Support**
Test Set				
Class 0 (Rabbit)	0.70	0.74	0.72	2755
Class 1 (Bobcat)	0.63	0.67	0.65	2209
Class 2 (Cat)	0.67	0.56	0.61	1737
Validation Set				
Class 0 (Rabbit)	0.77	0.88	0.82	1003
Class 1 (Bobcat)	0.91	0.80	0.85	900
Class 2 (Cat)	0.83	0.80	0.81	791
Bright Set				
Class 0 (Rabbit)	0.44	0.93	0.60	551
Class 1 (Bobcat)	0.69	0.25	0.37	442
Class 2 (Cat)	0.53	0.05	0.09	348

Table 4: BrightNet Per Class Performance

All Model Average Performance				
	Avg. Precision	**Avg. Recall**	**Avg. F1-Score**	**Accuracy**
Test Set				
ResNet-18	0.75	0.66	0.65	0.66
ResNet+	0.66	0.64	0.65	0.66
BrightNet	0.67	0.66	0.66	0.67
Validation Set				
ResNet-18	0.94	0.94	0.94	0.94
ResNet+	0.86	0.85	0.85	0.85
BrightNet	0.84	0.83	0.83	0.83
Bright Set				
ResNet-18	0.54	0.46	0.43	0.47
ResNet+	0.47	0.37	0.31	0.44
BrightNet	0.55	0.41	0.35	0.48

Table 5: Average Performance Across All Models

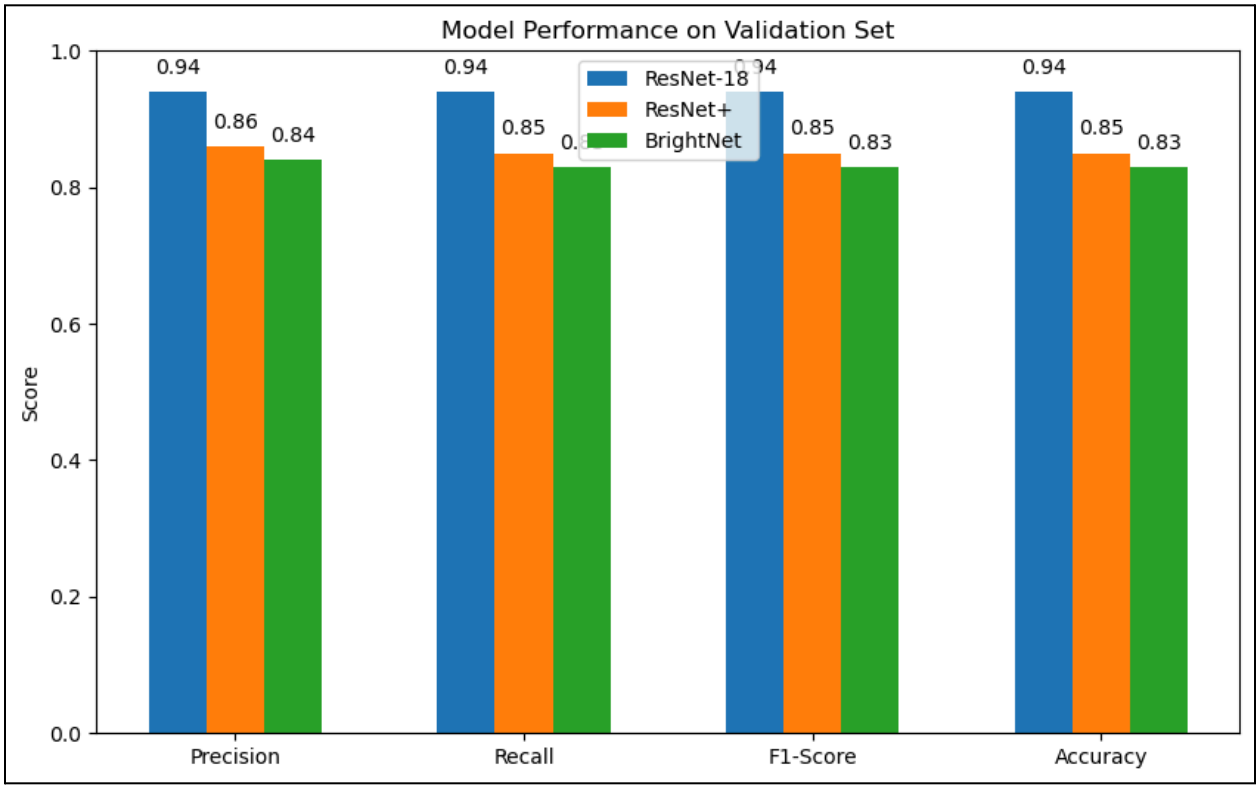


Chart 4: Model Performance on Validation Set

Appendix B

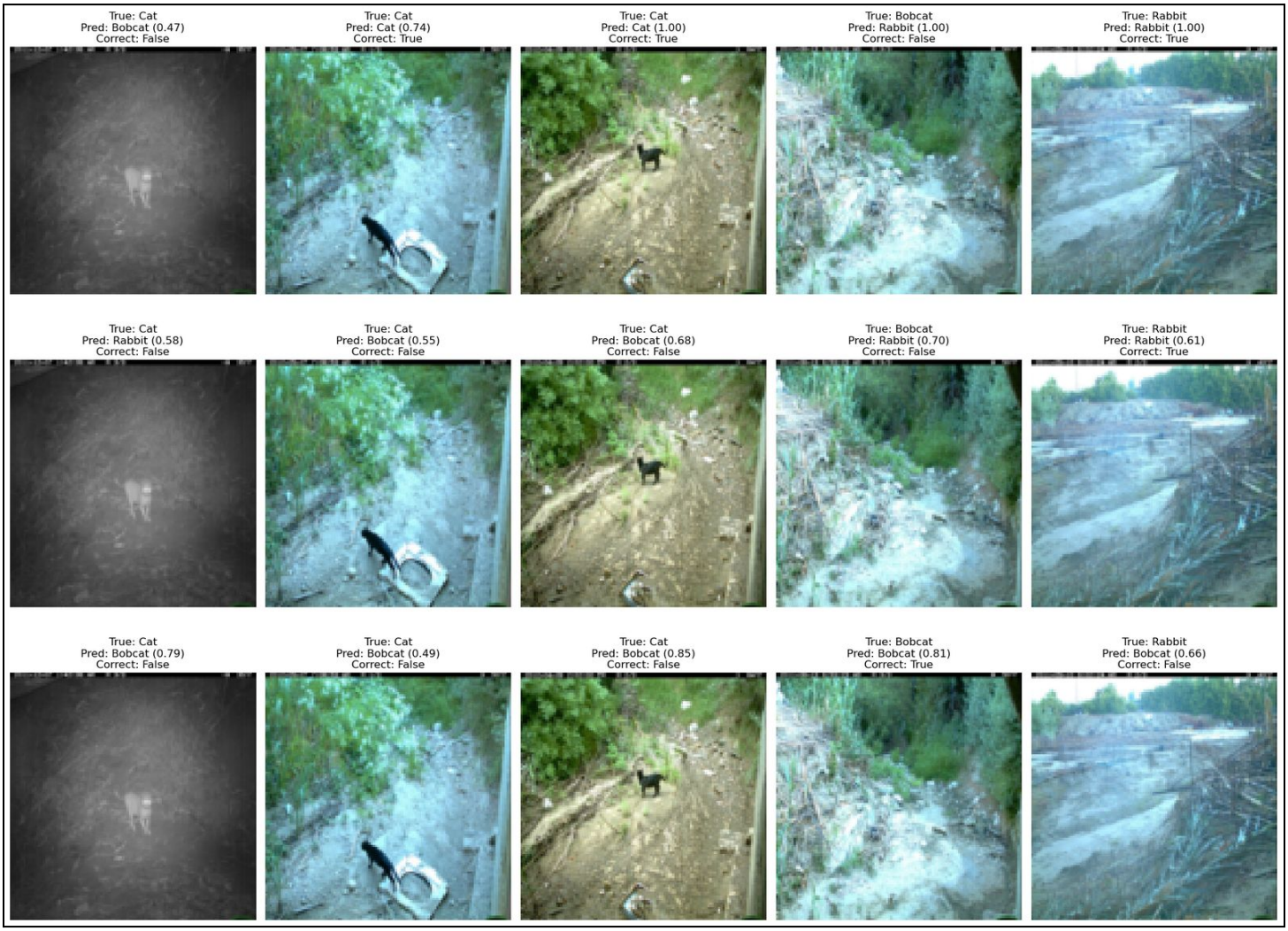


Figure 10: Predictions on Test Set

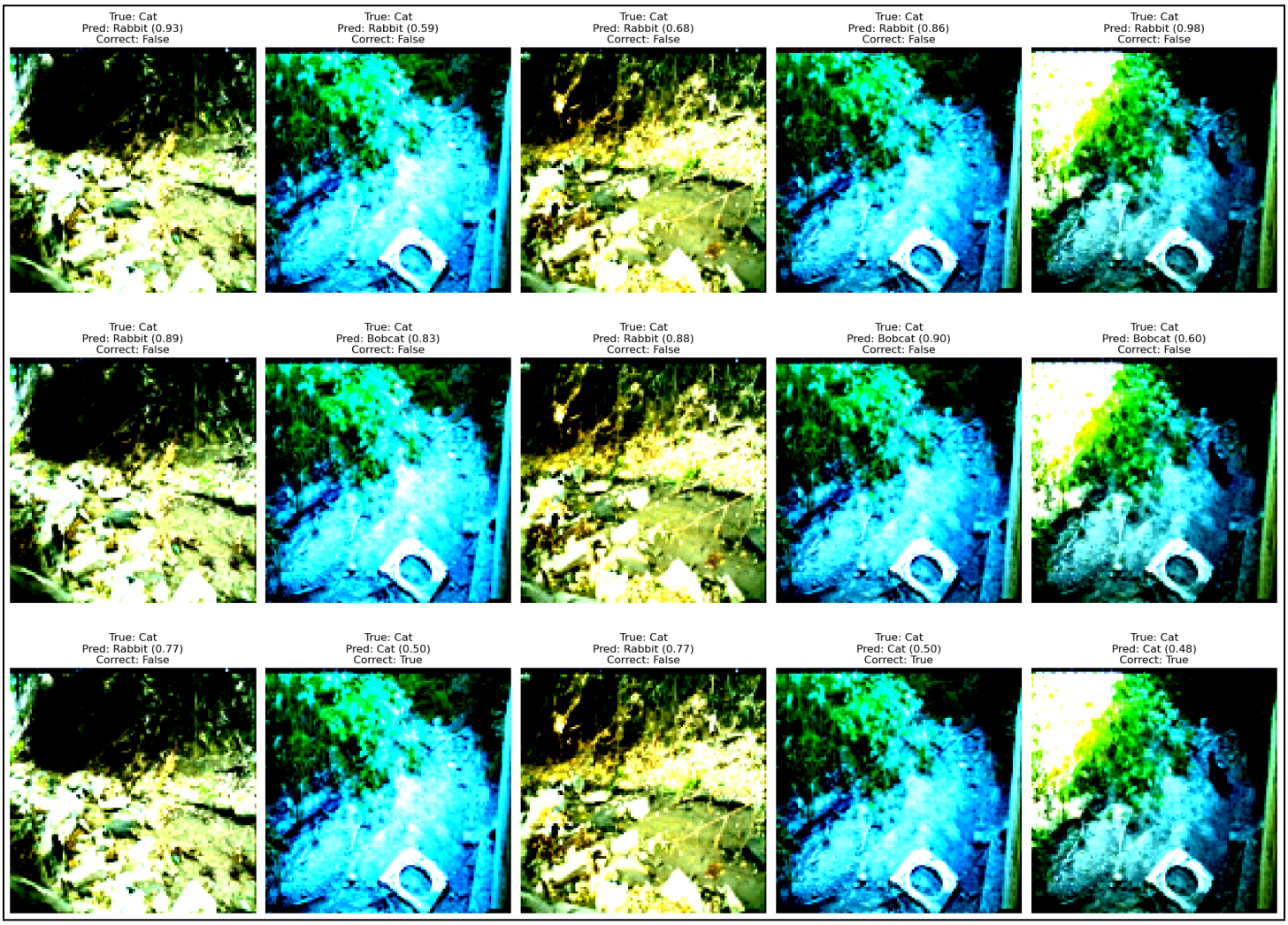


Figure 11: Predictions on Bright Set

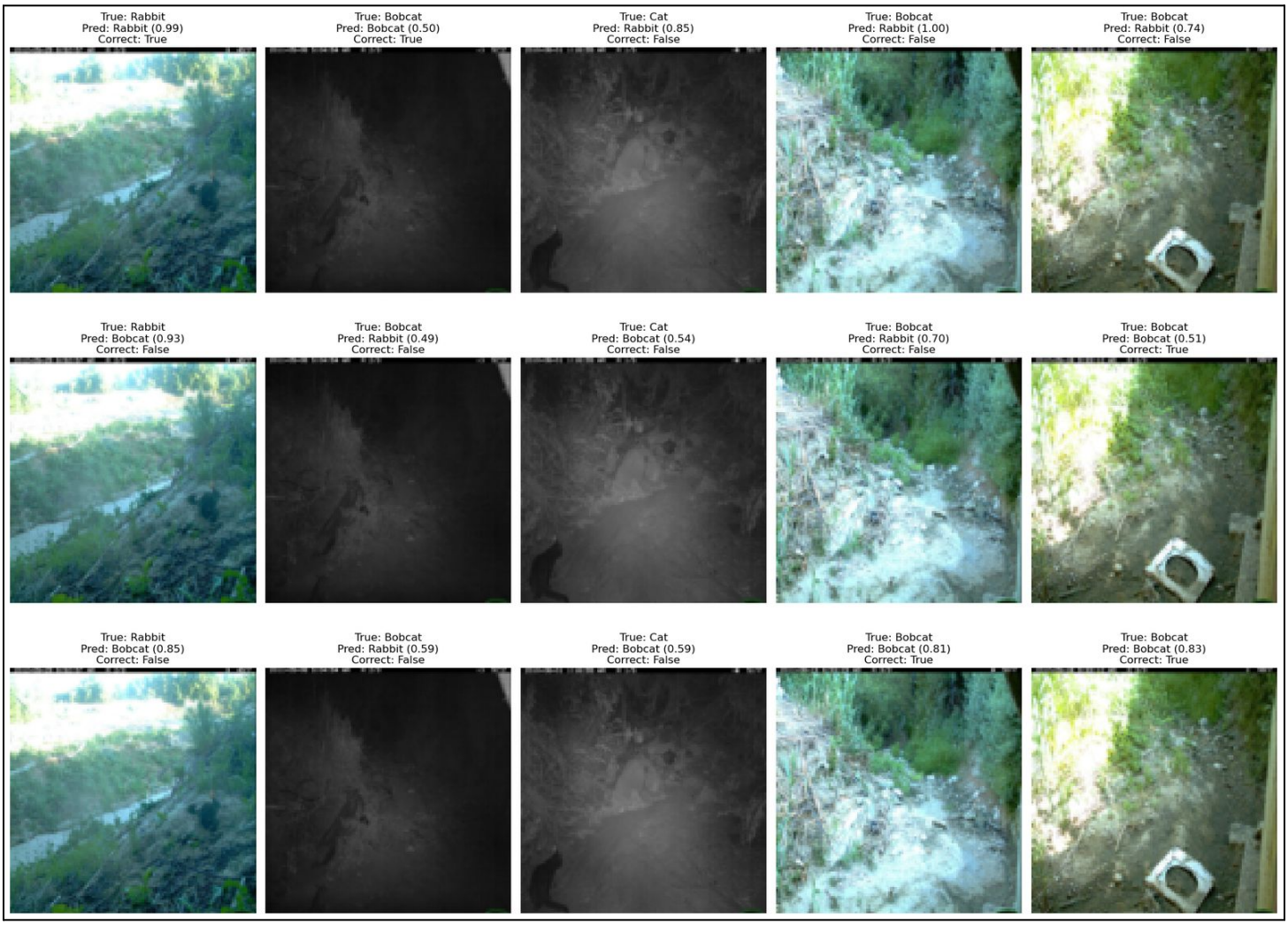


Figure 14: Additional Predictions on Test Set

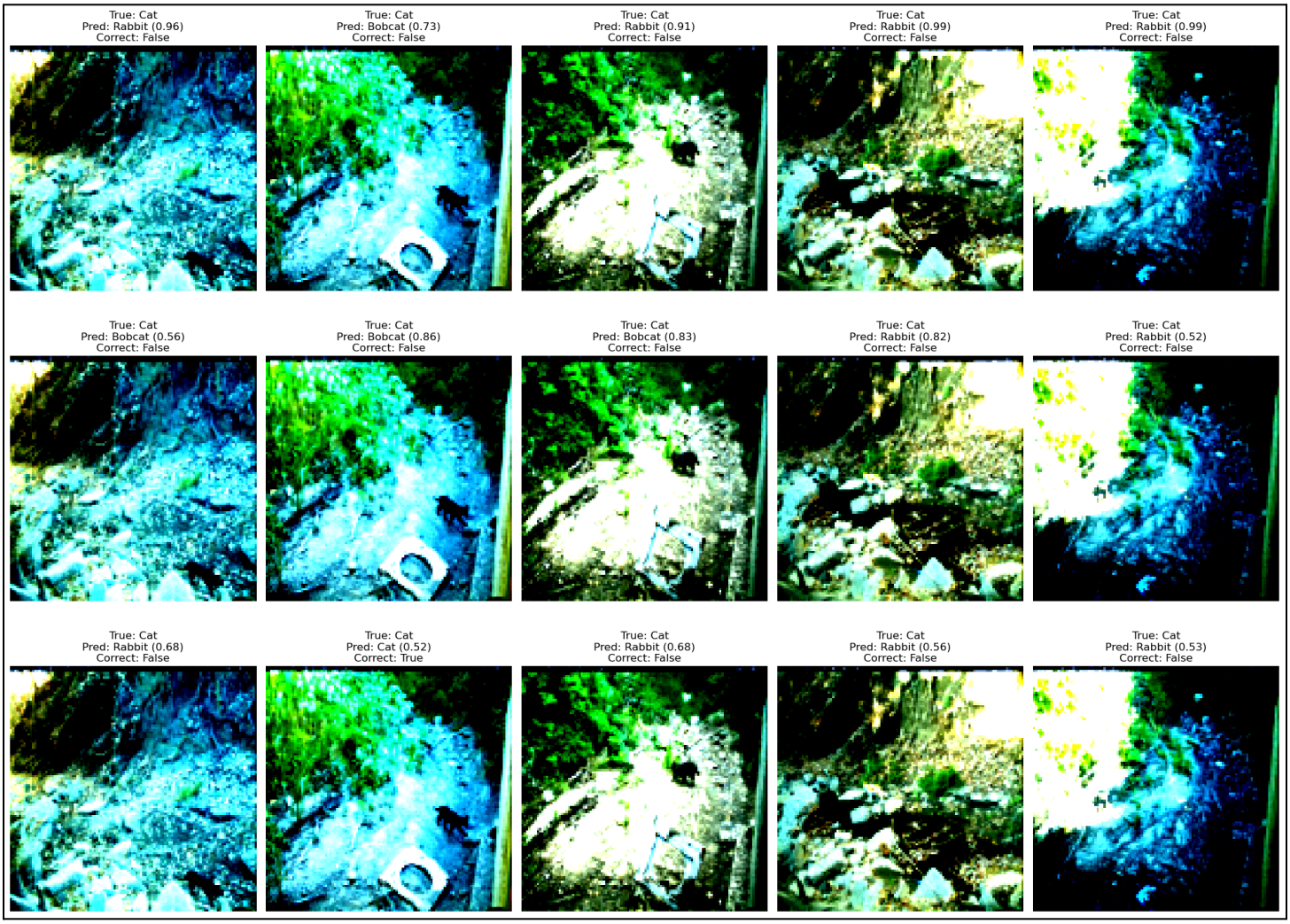


Figure 15: Additional Predictions on Bright Set

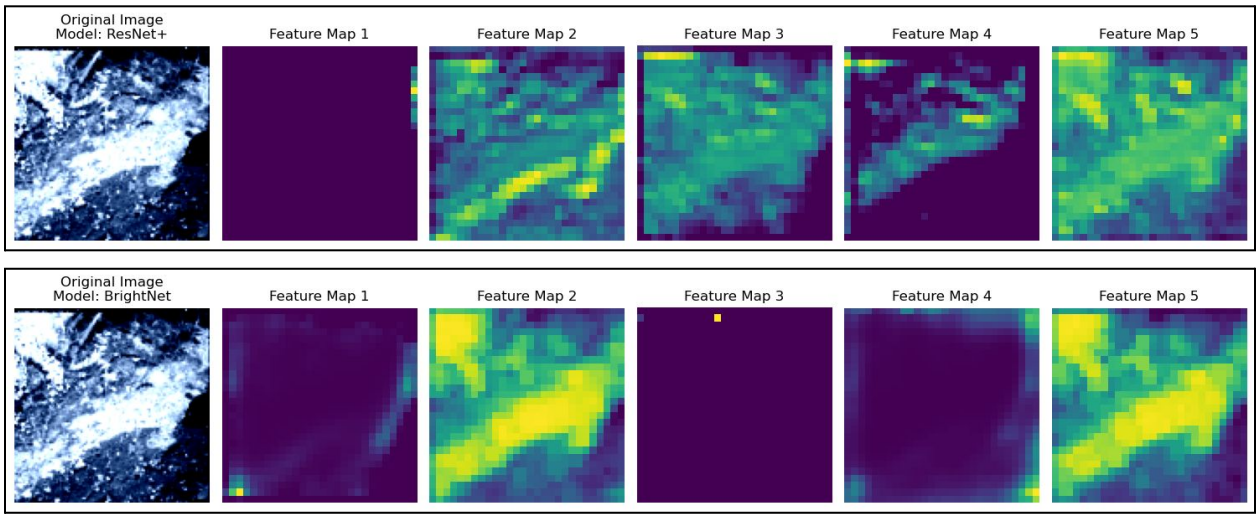


Figure 16: Additional Feature Maps for ResNet+ and BrightNet

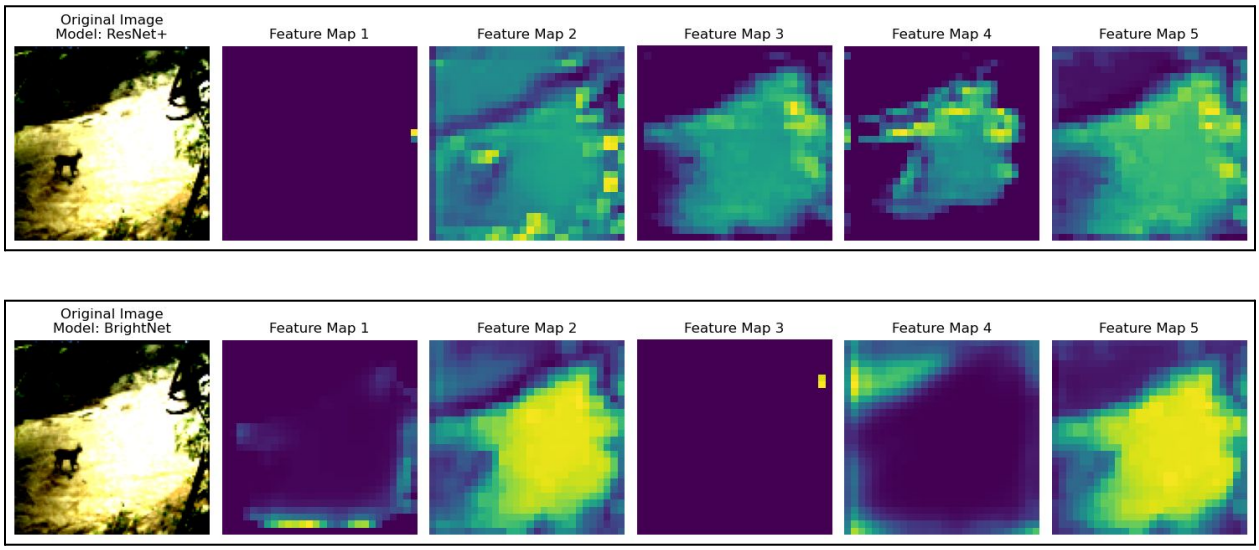


Figure 17: Additional Feature Maps for ResNet+ and BrightNet